TESS planet candidate around TIC 89062142

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Summary

Signal found by the bls search of a public-data TESS pipeline (SPOC 120 s PDCSAP light curves, sectors 105, 106, 107). Final tier: **candidate: likely planet**. Catalogue cross-match (TOIs, CTOIs, published planets, SPOC TCEs, TESS eclipsing binaries): **TOI 4674.01** (TOI). CNN score (calibrated): 0.916. TRICERATOPS (10 Monte Carlo runs, mean ± error of the mean): FPP $= 0.335 \pm 0.0052$, NFPP $= 7.2e-09$.

Host star (TIC v8)

RA, Dec (deg)	54.456978, -24.612603
TESS magnitude	13.2
Radius ($R_\odot$)	1.53 ± –
$T_{\rm eff}$ (K)	5595 ± 122
Density ($\rho_\odot$)	0.278 ± –
Distance (pc)	861

Transit fit

Limb-darkened model (quadratic law, Kipping q_1, q_2; GPU float64, 1024 annuli, 1024x4096 tables, 15x supersampling), emcee with 48 walkers, 8000 steps (748 burn-in), run until the chain exceeded 50 autocorrelation times. Medians and 68% intervals.

Mid-transit time T_0 ($\mathrm{BJD_{TDB}}$)	$2461240.9832^{+0.0010}_{-0.0010}$
Period P (days)	$4.24307^{+0.00017}_{-0.00017}$
Radius ratio $R_p/R_\star$	$0.0807^{+0.0022}_{-0.0022}$
Impact parameter b	$0.40^{+0.21}_{-0.25}$
Stellar density $\rho_\star$ (g cm^{-3})	$0.63^{+0.16}_{-0.21}$
Limb darkening q_1	$0.22^{+0.28}_{-0.14}$
Limb darkening q_2	$0.26^{+0.33}_{-0.18}$
Depth (ppm)	7211^{+257}_{-246}
Duration T_{14} (hours)	$3.87^{+0.10}_{-0.09}$
$a/R_\star$	$8.4^{+0.6}_{-1.1}$
Inclination (deg)	$87.3^{+1.7}_{-2.0}$
Planet radius ($R_\oplus$)	$13.5^{+1.0}_{-1.0}$
Semi-major axis (au)	$0.0592^{+0.0069}_{-0.0076}$
Equilibrium temperature (K, $A = 0$)	1365^{+94}_{-59}
Insolation ($S_\oplus$)	578^{+177}_{-93}

Vetting

Test	Value	Pass condition	Result
Transit SNR (correlated noise)	29.2	≥ 7.1	pass
Transits observed	11	≥ 2 (periodic)	pass
Odd/even depth difference (σ)	0.886	≤ 4	pass
Secondary eclipse (σ)	2.5	≤ 3.6	pass
Ingress fraction (V shape)	0.0828	≤ 0.4	pass
Planet radius ($R_\oplus$)	14	≤ 25	pass
Centroid source offset (arcsec)	0.865	4σ and $>21''$	pass
Difference image offset (arcsec)	1.87	4σ and $>10.5''$	pass
Difference image significance (σ)	0.47	–	–
CCD common mode (σ)	-0.0186	≤ 4	pass
Crowding (CROWDSAP)	0.96	≥ 0.8	pass
Stellar density ratio (max/TIC)	2.35	≥ 0.2	pass

Caveats

- This is a candidate with statistical evidence, not a confirmed planet: confirmation needs high-resolution imaging, reconnaissance spectroscopy and radial velocities.
- TRICERATOPS was run without a high-resolution imaging contrast curve, so close bound companions are not excluded.
- The MCMC converged (chain longer than 50 autocorrelation times, stable estimate) (acceptance 0.18, 8000 steps, about 2318 independent samples); errors are scaled by the time-averaging factor $\beta = 1.15$ for correlated noise.
- Stellar radius and effective temperature come from TIC v8; if the TIC gives no error, 7% (radius) and 150 K were assumed.

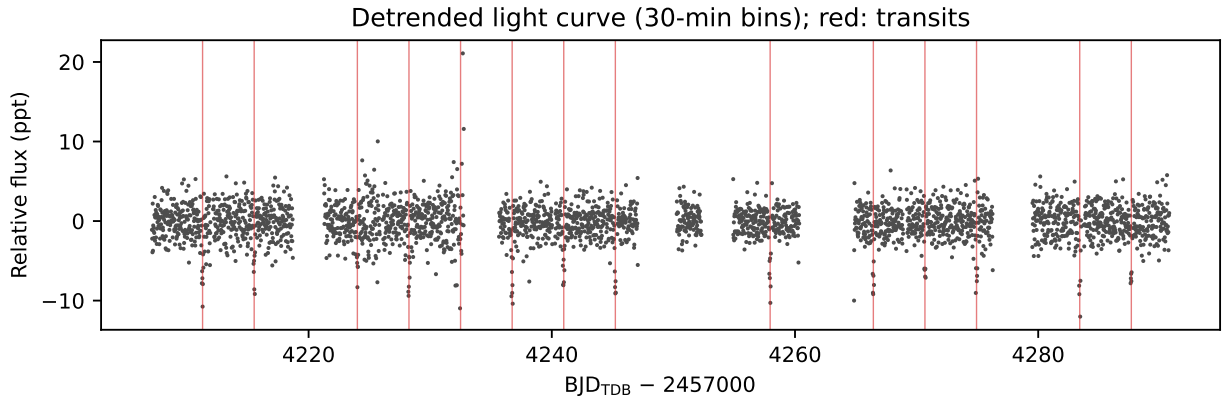


Figure 1: Detrended SPOC PDCSAP light curve (transit-protected biweight).

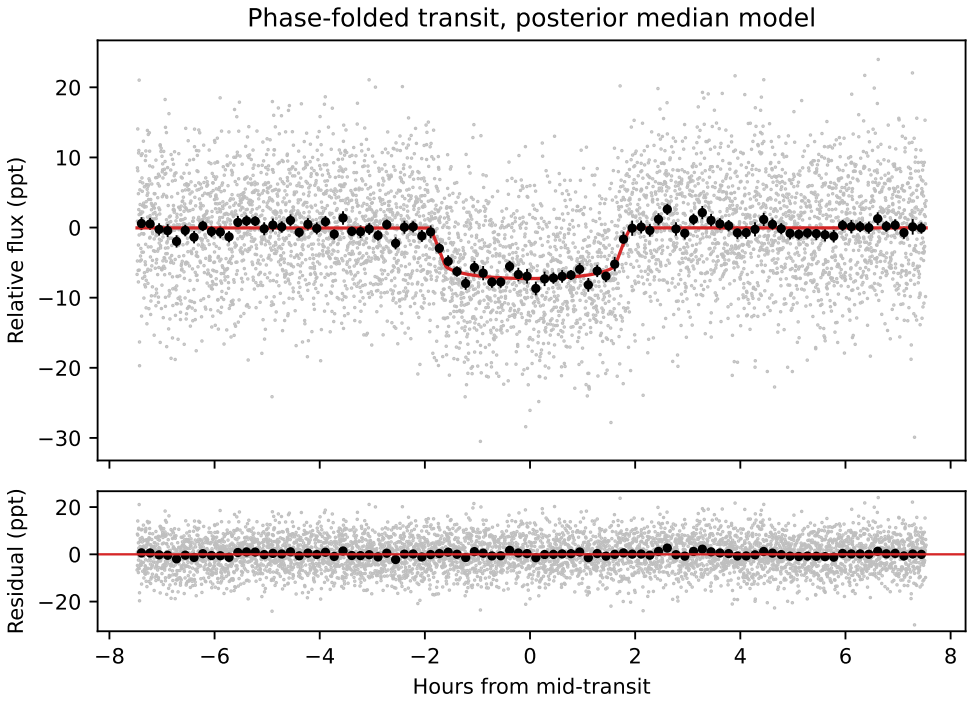


Figure 2: Phase-folded transit with the posterior median limb-darkened model.

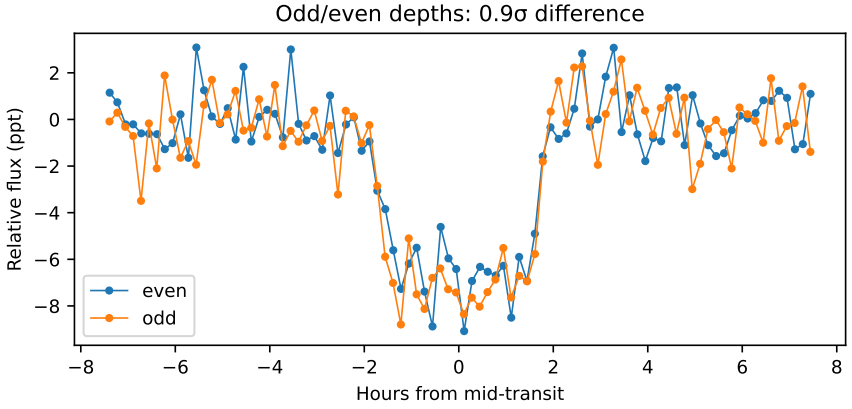


Figure 3: Odd and even transits, binned to 10 minutes.

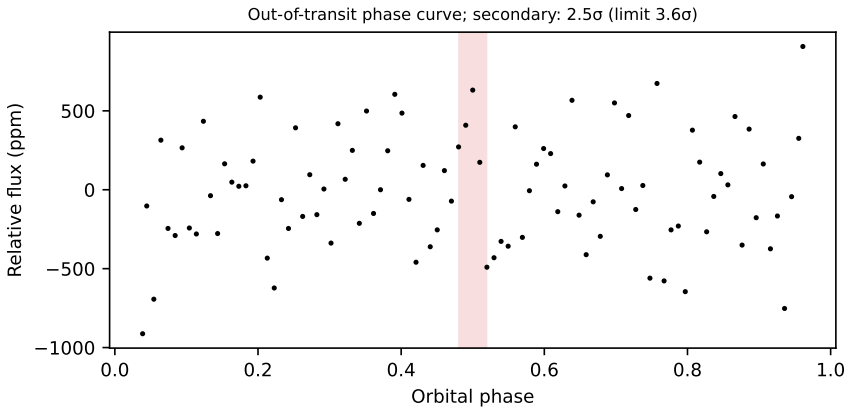


Figure 4: Phase curve; the shaded band is the secondary eclipse window for a circular orbit.

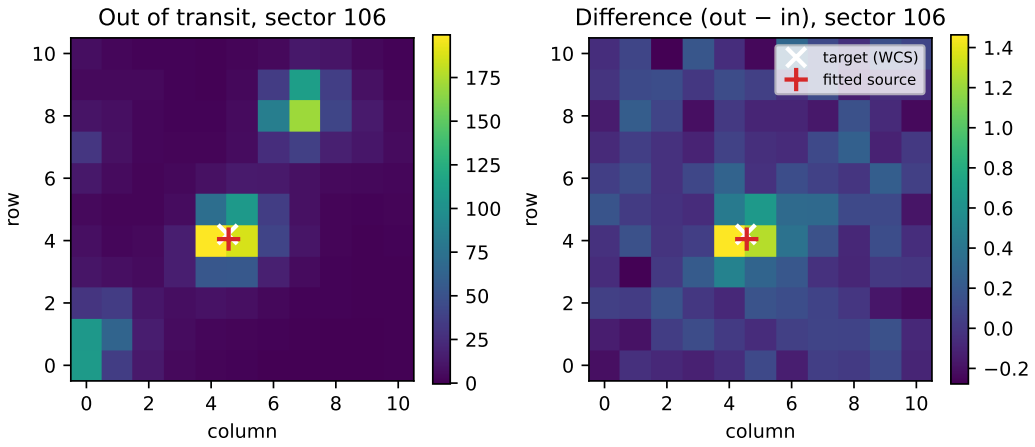


Figure 5: SPOC target pixel file: out-of-transit image and difference image. White cross: target catalogue position; red plus: fitted source.

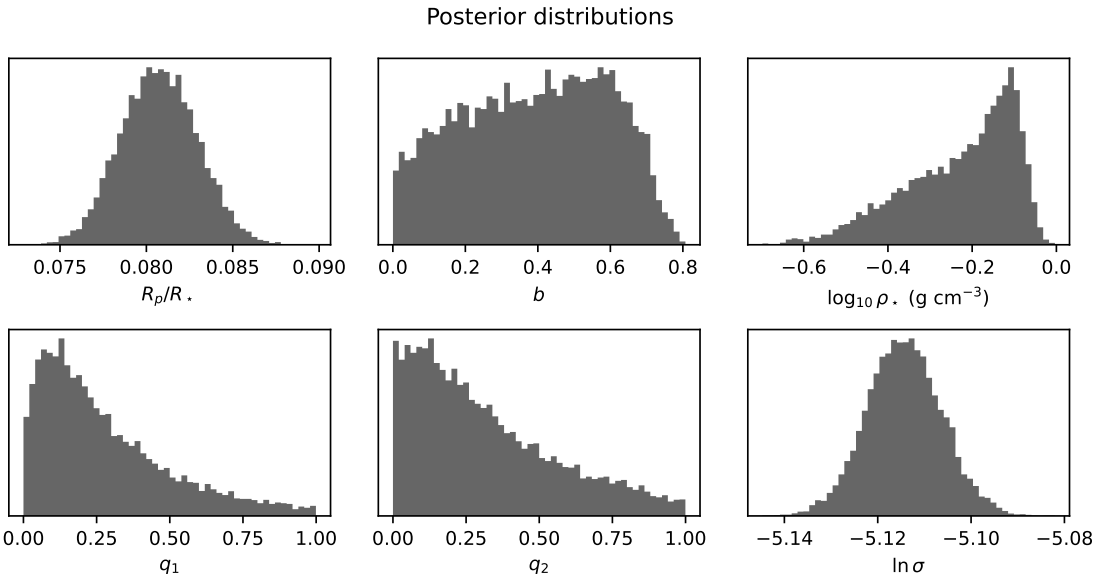


Figure 6: Posterior distributions of the fitted parameters.